Ubuntu Machine — Debugging & Fix Reference

Machine: sjay | Kernel: 6.8.0-117-generic (Ubuntu 24.04) | Compiled: May 2026

1. Problem Overview

This machine (AMD/Intel platform, ASMedia AHCI controller at 0000:02:00.1, RTL8168h NIC at 0000:04:00.0) exhibited three interconnected failure modes all sharing the same root cause: **PCIe/SATA power management (ASPM + LPM) incompatible with the installed hardware**. The issues manifested as disk dropouts after suspend/resume, an unresponsive ethernet NIC, and persistent SATA PHY signal errors.

Issue	Root Cause	Fix Applied
SATA disks dropping on resume	PCIe ASPM + SATA LPM sending drives to power states they can't recover from	pcie_aspm=off, libata.force=noncq, LPM udev rule
Resume failure (ACPI)	BIOS not saving/restoring non-volatile sleep state (S3)	acpi_sleep=nonvs kernel parameter
RTL8168h NIC dying after ~17h uptime	PCIe ASPM freezing NIC chip registers, firmware reload fails	pcie_aspm=off kernel parameter
sda1 EXT4 journal aborts	Drive going offline mid-write due to PHY drops	All above + SATA cable replacement
grub fixes not persisting	/etc/default/grub.d/sleep-fix.conf overriding main grub config	Merged all params into sleep-fix.conf and /etc/default/grub
sda (ADATA SU635) continuous PHY drops	Marginal/failing SATA cable (UnrecovData + BadCRC + DevExch SErrors)	Replace SATA cable; ddrescue backup in progress

2. Kernel Boot Parameters

The final working kernel cmdline (in both **/etc/default/grub** and **/etc/default/grub.d/sleep-fix.conf**):

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash acpi_sleep=nonvs libata.force=noncq pcie_aspm=off mem_sleep_default=s2idle"
```

Apply with:

```
sudo update-grub sudo reboot
```

Verify after reboot:

```
cat /proc/cmdline
```

Parameter Explanations

Parameter	Purpose
`acpi_sleep=nonvs`	Prevents BIOS from saving/restoring non-volatile sleep state during S3. Fixes resume failures where PCIe devices don't come back after suspend.
`libata.force=noncq`	Disables Native Command Queuing on all ATA devices. Prevents the kernel from queuing multiple commands to drives that can't reliably handle NCQ after a PHY error recovery.
`pcie_aspm=off`	Disables PCIe Active State Power Management system-wide. Prevents the PCIe controller from putting devices (NIC, AHCI) into L1 sleep states they can't wake from cleanly.
`mem_sleep_default=s2idle`	Forces suspend-to-idle (S0ix) sleep mode instead of S3. More compatible with modern AMD/Intel platforms where S3 causes PCIe re-enumeration issues.

3. SATA Link Power Management (LPM) Fix

The AHCI controller (flags include **apst** — Aggressive Power State Transitions) autonomously power-cycles the SATA PHY, causing both the ADATA SU635 (sda) and Hitachi HUA723020ALA641 (sdb) to fail link re-negotiation. The fix forces maximum performance mode.

udev Rule — /etc/udev/rules.d/50-sata-lpm.rules

```
ACTION=="add", SUBSYSTEM=="scsi_host", KERNEL=="host*",
ATTR{link_power_management_policy}="max_performance"
```

Apply without rebooting:

```
sudo udevadm control --reload-rules sudo udevadm trigger --subsystem-match=scsi_host #
Verify: cat /sys/class/scsi_host/host*/link_power_management_policy # All lines must read:
max_performance
```

Note: If output shows *keep_firmware_settings*, the rule has not fired. Force it immediately with:

```
echo max_performance | sudo tee /sys/class/scsi_host/host*/link_power_management_policy
```

4. Grub Configuration — Important Gotcha

Files in **/etc/default/grub.d/** are sourced *after* /etc/default/grub, meaning the last GRUB_CMDLINE_LINUX_DEFAULT assignment wins. The machine had **sleep-fix.conf** in that directory which was silently overriding all parameters set in /etc/default/grub.

Resolution

Keep parameters consistent in both files, or remove grub.d snippets and use only /etc/default/grub.

Verify the compiled grub config actually contains your parameters before rebooting:

```
sudo grep 'pcie_aspm\|acpi_sleep\|mem_sleep' /boot/grub/grub.cfg | head -3 # Must show all
parameters in the linux boot line
```

5. ASPM Verification

After applying pcie_aspm=off and rebooting, verify no PCIe links have ASPM enabled:

```
sudo lspci -vv | grep -i 'LnkCtl:' | grep -v 'Disabled' # Should return only
LnkCtl2/LnkCtl3 lines (speed/compliance settings) # Any 'ASPM L1 Enabled' lines indicate
the fix did not take effect
```

6. Disk Health Status

sda — ADATA SU635 (240 GB, ata1/host0)

SMART health: **PASSED**. Zero reallocated sectors, zero CRC errors, no errors logged. Despite clean SMART, drive is exhibiting continuous PHY-level SErrors with DevExch (device exchange — drive briefly disappearing from bus). This is characteristic of a **failing or loose SATA cable**, not internal drive failure.

Current status: Active PHY drops as of May 25 2026. ddrescue backup in progress.

SError pattern observed:

```
SError: { RecovComm UnrecovData Proto HostInt PHYRdyChg PHYInt CommWake 10B8B BadCRC Handshk DevExch }
```

The escalating presence of **UnrecovData**, **BadCRC**, and **Handshk** errors confirms deteriorating signal integrity — consistent with a marginal cable, not a power management issue.

sdb — Hitachi Ultrastar HUA723020ALA641 (2 TB, ata6/host5)

SMART health: **PASSED**. After 22,090 power-on hours: zero reallocated sectors, zero pending sectors, zero uncorrectable, zero UDMA CRC errors. Occasional IDENTIFY failures on resume — likely a marginal SATA cable on ata6 port.

Recommended action: Replace SATA cable on ata6 port as a precaution.

---

7. Data Recovery — ADATA SU635 (sda)

Due to active PHY drops causing I/O errors on /Data (sda1), rsync failed with readdir I/O error (errno 5). Use ddrescue for recovery:

```
sudo apt install gddrescue sudo umount /Data sudo ddrescue -d -r3 /dev/sda1 /BackupDrive/sda1_rescue.img /BackupDrive/sda1_rescue.log
```

If the drive disconnects mid-run, resume with:

```
sudo ddrescue -d -r3 -R /dev/sda1 /BackupDrive/sda1_rescue.img /BackupDrive/sda1_rescue.log
```

Mount the rescued image:

```
sudo mkdir /mnt/rescued sudo mount -o loop /BackupDrive/sda1_rescue.img /mnt/rescued
```

---

8. RTL8168h Ethernet NIC Fix

The RTL8168h (enp4s0) was dying after approximately 17 hours uptime. The failure sequence: NETDEV WATCHDOG transmit queue timeout → chip registers unresponsive (rtl_chipcmd_cond, rtl_ephyar_cond, rtl_eriar_cond all timing out) → firmware reload fails with error -110 (ETIMEDOUT). Root cause: PCIe ASPM placing the chip in a power state it cannot recover from.

The **pcie_aspm=off** kernel parameter resolves this. Additionally, remove any invalid r8169 module parameters:

```
# Check for invalid parameters: cat /etc/modprobe.d/*.conf | grep r8169 # If 'options r8169 aspm=...' found, remove that file: sudo rm /etc/modprobe.d/r8169.conf
```

Note: The 'aspm' parameter is not valid for the r8169 driver and is silently ignored. Only the kernel-level pcie_aspm=off flag is effective.

---

9. Filesystem Recovery

Multiple journal aborts occurred on sda1 due to drive going offline mid-write. Run fsck from a live USB (do not run on a mounted filesystem):

```
# Boot from Ubuntu live USB, then: sudo fsck -f -y /dev/sda1 sudo fsck -f -y /dev/sdb1
```

EXT4 recovery on mount ('recovery complete') handles minor journal issues automatically, but a full fsck is recommended after any journal abort to check for inode/directory corruption.

10. Recommended Next Steps

1	Complete ddrescue backup of sda1 to /BackupDrive before any physical intervention
2	Power off and replace the SATA data cable on ata1 (sda port)
3	Replace SATA data cable on ata6 (sdb port) as precaution
4	Move sda to a different SATA port on the motherboard to rule out port fault
5	After reassembly, monitor for SErrors: sudo dmesg | grep 'SError\\|DevExch\\|disable device'
6	If PHY drops persist on new cable and different port, add a PCIe SATA card (~$15) to bypass the onboard AHCI port
7	Run smartctl -t long /dev/sda after stable operation to confirm drive health
8	Consider replacing ADATA SU635 with a more reliable SSD (Samsung 870 EVO, Crucial MX500) — budget QLC NAND has poor power rail tolerance